THE INDIAN INTERNATIONAL SCHOOL,DSO
Worksheet – Chapter 7: Integrals (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	$\int e^x(\sin x + \cos x)\,dx$ is equal to: (A) $e^x\cos x + C$ (B) $e^x\sin x + C$ (C) $-e^x\sin x + C$ (D) $e^x(\sin x - \cos x) + C$	1
2.	If $\int \frac{2^{1/x}}{x^2}\,dx = k \cdot 2^{1/x} + C$, then k is equal to: (A) $-\frac{1}{\log 2}$ (B) $-\log 2$ (C) -1 (D) $\frac{1}{2}$	1
3.	The value of $\int_1^e \log x\,dx$ is: (A) 0 (B) 1 (C) e (D) $e\log e$	1
4.	$\int 2^{x+2}\,dx$ is equal to: (A) $2^{x+2} + C$ (B) $2^{x+2}\log 2 + C$ (C) $\frac{2^{x+2}}{\log 2} + C$ (D) $2 \cdot \frac{2^x}{\log 2} + C$	1
5.	If $\int_0^1 \frac{e^x}{1+x}\,dx = \alpha$, then $\int_0^1 \frac{e^x}{(1+x)^2}\,dx$ is equal to: (A) $\alpha - 1 + \frac{e}{2}$ (B) $\alpha + 1 - \frac{e}{2}$ (C) $\alpha - 1 - \frac{e}{2}$ (D) $\alpha + 1 + \frac{e}{2}$	1

Q No.	Question	Marks
6.	**Assertion (A):** $\int_{-1}^{1} x\,\|x\|\,dx = 0$. **Reason (R):** $f(x) = x\|x\|$ is an odd function, and $\int_{-a}^{a} f\,(x)\,dx = 0$ for every odd function f. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Find $\int \frac{x^3-x^2+x-1}{x-1}\,dx$.	2
	LEVEL 2 **Q 8 – Q 14**	**19**
8.	Evaluate: $\int_0^{\pi/2} \sin 2x \cos 3x\,dx$.	2
9.	Given $\frac{d}{dx}F(x) = \frac{1}{\sqrt{2x-x^2}}$ and $F(1) = 0$, find $F(x)$.	2
10.	Find: $\int \frac{2x-1}{(x-1)(x+2)(x-3)}\,dx$.	3
11.	Evaluate $\int_0^{\pi/4} \log(1+\tan x)\,dx$.	3
12.	Find $\int e^x \left(\frac{1+\sin x}{1+\cos x}\right) dx$.	3
13.	Evaluate: $\int_0^5 (\|x-1\| + \|x-2\| + \|x-5\|)\,dx$.	3
14.	Find $\int \sin^3 x \cos^2 x\,dx$.	3
	LEVEL 3 **Q 15 – Q 20**	**27**
15.	Evaluate $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\,dx$.	5
16.	Find $\int \frac{x^2+1}{(x-1)^2(x+3)}\,dx$.	5
17.	Evaluate $\int_0^{\pi/2} \log \sin x\,dx$.	5
18.	Find $\int \frac{x+2}{\sqrt{x^2+2x+3}}\,dx$.	3
19.	Find $\int \left(\sqrt{\tan x} + \sqrt{\cot x}\right) dx$.	5
20.	**Case Study:** In class, a teacher demonstrates the property $\int_0^a f\,(x)\,dx = \int_0^a f\,(a-x)\,dx$, which makes many definite integrals easy to evaluate. **(i)** Evaluate $\int_0^{\pi/2} \frac{\cos x}{\sin x+\cos x}\,dx$. **[1]** **(ii)** Evaluate $\int_0^{\pi/2} \frac{\sin^n x}{\sin^n x+\cos^n x}\,dx$. **[1]** **(iii)** Evaluate $\int_0^1 x\,(1-x)^{99}\,dx$. **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 7: Integrals

Q No.	Answer / Key Steps
	LEVEL 1
1.	(B) form $e^x[f(x)+f'(x)]$ with $f=\sin x$
2.	(A) put $t=\frac{1}{x}$: $-\frac{2^{1/x}}{\log 2}+C$
3.	(B) $[x\log x-x]_1^e=1$
4.	(C) $\frac{2^{x+2}}{\log 2}+C$
5.	(B) $\int_0^1 e^x\left[\frac{1}{1+x}-\frac{1}{(1+x)^2}\right]dx=\left[\frac{e^x}{1+x}\right]_0^1=\frac{e}{2}-1$, so the integral $=\alpha+1-\frac{e}{2}$
6.	(A) Both true; R explains A.
7.	$\int(x^2+1)\,dx=\frac{x^3}{3}+x+C$
	LEVEL 2
8.	$\frac{1}{2}\int_0^{\pi/2}(\sin 5x-\sin x)\,dx=\frac{1}{2}\left[-\frac{\cos 5x}{5}+\cos x\right]_0^{\pi/2}=-\frac{2}{5}$
9.	$[2x-x^2=1-(x-1)^2$, so $F(x)=\sin^{-1}(x-1)+C$; $F(1)=0\Rightarrow C=0$; $F(x)=\sin^{-1}(x-1)$
10.	$-\frac{1}{6}\log\lvert x-1\rvert-\frac{1}{3}\log\lvert x+2\rvert+\frac{1}{2}\log\lvert x-3\rvert+C$
11.	Using $x\to\frac{\pi}{4}-x$: $2I=\frac{\pi}{4}\log 2\Rightarrow I=\frac{\pi}{8}\log 2$
12.	Integrand $=e^x\left(\tan\frac{x}{2}+\frac{1}{2}\sec^2\frac{x}{2}\right)$; answer $e^x\tan\frac{x}{2}+C$
13.	$\frac{17}{2}+\frac{13}{2}+\frac{25}{2}=\frac{55}{2}$
14.	Put $t=\cos x$: $-\frac{\cos^3 x}{3}+\frac{\cos^5 x}{5}+C$
	LEVEL 3
15.	Using $x\to\pi-x$: $2I=\pi\int_0^{\pi}\frac{\sin x}{1+\cos^2 x}dx=\pi\cdot\frac{\pi}{2}\Rightarrow I=\frac{\pi^2}{4}$
16.	$\frac{3}{8}\log\lvert x-1\rvert-\frac{1}{2(x-1)}+\frac{5}{8}\log\lvert x+3\rvert+C$
17.	$2I=\int_0^{\pi/2}\log\left(\frac{\sin 2x}{2}\right)dx=I-\frac{\pi}{2}\log 2\Rightarrow I=-\frac{\pi}{2}\log 2$
18.	$x+2=\frac{1}{2}(2x+2)+1$: $\sqrt{x^2+2x+3}+\log\left\lvert x+1+\sqrt{x^2+2x+3}\right\rvert+C$
19.	Integrand $=\frac{\sin x+\cos x}{\sqrt{\sin x\cos x}}$; put $t=\sin x-\cos x$: $\sqrt{2}\sin^{-1}(\sin x-\cos x)+C$
20.	(i) $\frac{\pi}{4}$. (ii) $\frac{\pi}{4}$. (iii) $\int_0^1(1-x)\,x^{99}dx=\frac{1}{100}-\frac{1}{101}=\frac{1}{10100}$